

Place Value

Use the cue, then fade the cue. Study the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS Each place is **10× the place to its right**. From right to left the whole-number places are ones, tens, hundreds, thousands. To the right of the decimal point come tenths, hundredths, and thousandths. A digit's **value** equals the digit × its place. Example: in **3,472** the 4 is in the hundreds place, so its value is $4 \times 100 = 400$.

Round 1 — Retrieve It

1. Each place is how many times the place to its right?

2. In the number 3,472, what is the value of the 4?

3. Name the first three places to the right of the decimal point.

4. How do you find a digit's value?

Round 2 — Sketch It

Draw a place-value chart for the number 5,286. Label each column and write the digit and its value under each place.

Place Value — Apply It

Keep the cue card covered. Answer first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Idea

1. Write the value of each 6 in the number 6,619. Which 6 is worth more, and why?

2. Write the number 4.207 in words. Name the smallest place used.

3. A digit moves one place to the left. How does its value change, and why?

Round 4 — Reason from Evidence

Maria says the 7 in 7,000 is worth 100 times the 7 in 70. Is she correct? Use place value to explain how the two 7s compare.

Place Value — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain how the value of a digit depends on its place. Use the words digit, place, and value.

C**Connect**

Connect place value to money. Show how \$1.35 uses ones, tenths (dimes), and hundredths (pennies).

E**Extend**

Write a number that has a 9 in the hundreds place and a 9 in the tenths place. Tell the value of each 9.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Place Value

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. 10 times (each place is $10\times$ the place to its right).
2. 400 (the 4 is in the hundreds place: 4×100).
3. Tenths, hundredths, thousandths.
4. Multiply the digit by its place value (digit $\times$ place).

Sketch It — look for

Chart columns thousands, hundreds, tens, ones with digits 5, 2, 8, 6. Values shown as 5,000; 200; 80; 6.

Apply It

1. The first 6 is 6,000 and the second 6 is 600. The 6 in the thousands place is worth more (10 times as much).
2. Four and two hundred seven thousandths. The smallest place used is thousandths.
3. Its value becomes 10 times as large, because each place to the left is $10\times$ the place to its right.

Reason from Evidence — look for

Yes. The 7 in 7,000 is worth 7,000 and the 7 in 70 is worth 70. $7,000 \div 70 = 100$, so it is 100 times as much.

ACE — Extend look-for

Any number such as 900.9: the 9 in the hundreds place is worth 900 and the 9 in the tenths place is worth 0.9 (nine tenths).

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color.